UNIVERSITI KUALA LUMPUR

MALAYSIAN INSTITUTE OF INFORMATION TECHNOLOGY

L01-B04

LAB WEEK 5 :

FOOTPRINTING

BIL	NAME	MATRIX NUMBER
1	MUHAMMAD SYAKIRIN BIN MOHD HANIF	52215125427

INSTRUCTOR: NOR ADANI KAMAL MOHAMAD NASIR

Introduction

This laboratory report documents a comprehensive footprinting and reconnaissance for lab week 5 assessment. The objective is basically to perform passive and active reconnaissance on public, non-bank target. For this assessment, **Shell Malaysia (shell.com.my)** was selected as my target for this lab.

This report details a multi-step reconnaissance by utilizing a variety of industry-standard tools :

- Domain Footprinting : Using whois and ping to disclose domain details such as ownership and discover others details on IP address.

- Website Fingerprinting : Utilizing scanners such as VirusTotal to analyze domain reputation and ensure the target is not doing any malicious activity.

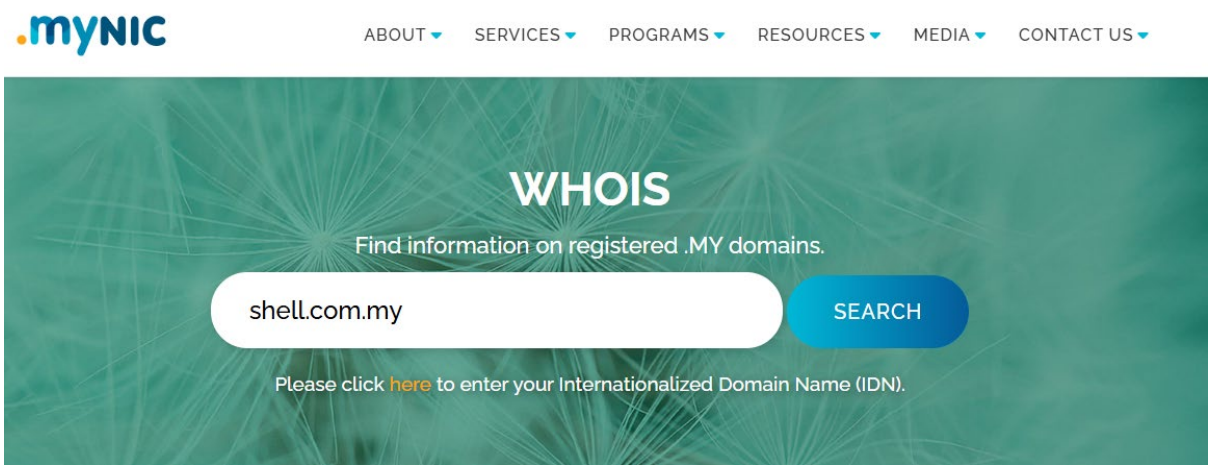
- Source-code Investigation : Exploring the Github dorking and identify why large corporates often utilize private repositories over public repositories.

- Reputation and Alerts : Monitoring the organization's online integrity and abuse history.

Methodology

❖ Steps 1 : *Domain Footprinting*

1) Whois



Domain Name	shell.com.my
Domain Registration No.	D10721-MYNIC
Domain Registration Date	01 April 1997 00:00:00 (Malaysia Time) 31 March 1997 16:00:00 (UTC)
Domain Expiry Date	01 April 2027 00:00:00 (Malaysia Time) 31 March 2027 16:00:00 (UTC)
Domain Last Updated	26 February 2026 22:50:26 (Malaysia Time) 26 February 2026 14:50:26 (UTC)
Domain Status	active
DNSSEC	Signed Delegation
Registrar	Webplus Digital Sdn Bhd.
Registrar Phone Number	tel:+603.89966788
Nameservers	dns1.shell.com dns2.shell.com dns3.shell.com

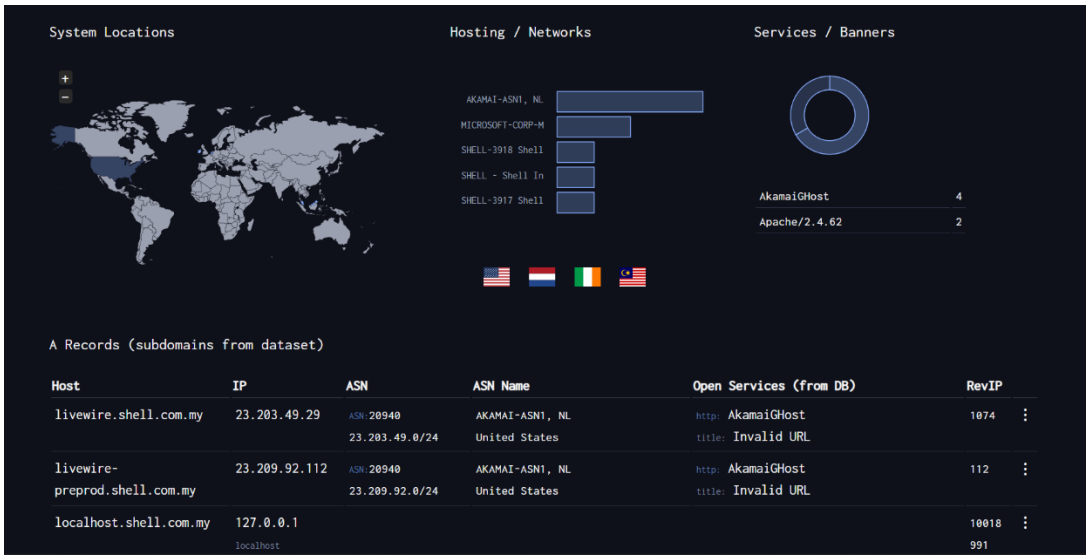
The **whois** results for shell.com.my is confirming the domain's legitimacy, have been registered since April 1997. It's managed by Webplus Digital Sdn Bhd, and secured with DNNSEC which is "Signed Delegation" to prevent DNS hijacking. Additionally, the use of dedicated servers of **dns1.shell.com** and **dns3.shell.com** indicates a robust DNS infrastructure.

2) Nslookup

```
┌──(kali㉿kali)-[~]
└─$ nslookup shell.com.my
Server:         192.168.0.1
Address:        192.168.0.1#53

Non-authoritative answer:
Name:   shell.com.my
Address: 4.210.156.184
```

The **nslookup** command for **shell.com.my** resulting non-authoritative answer. The term "non-authoritative" indicates the information retrieved from local DNS server rather than actual Shell's official website.

3) DNS Dumpster



The DNSDumpster results for **shell.com.my** provides a pure details of the target's infrastructure and hosting environment. This reveals that Shell's are operating digitally in United States, Netherlands, Ireland and Malaysia. The host section highlight Akamai and Microsoft Corporation infrastructure, and indicate the use of CDNs which are cloud services managing security and IT. Furthermore, the "Services/Banners" displays **AkamaiGHost** and **Apache/2.4.62** servers.

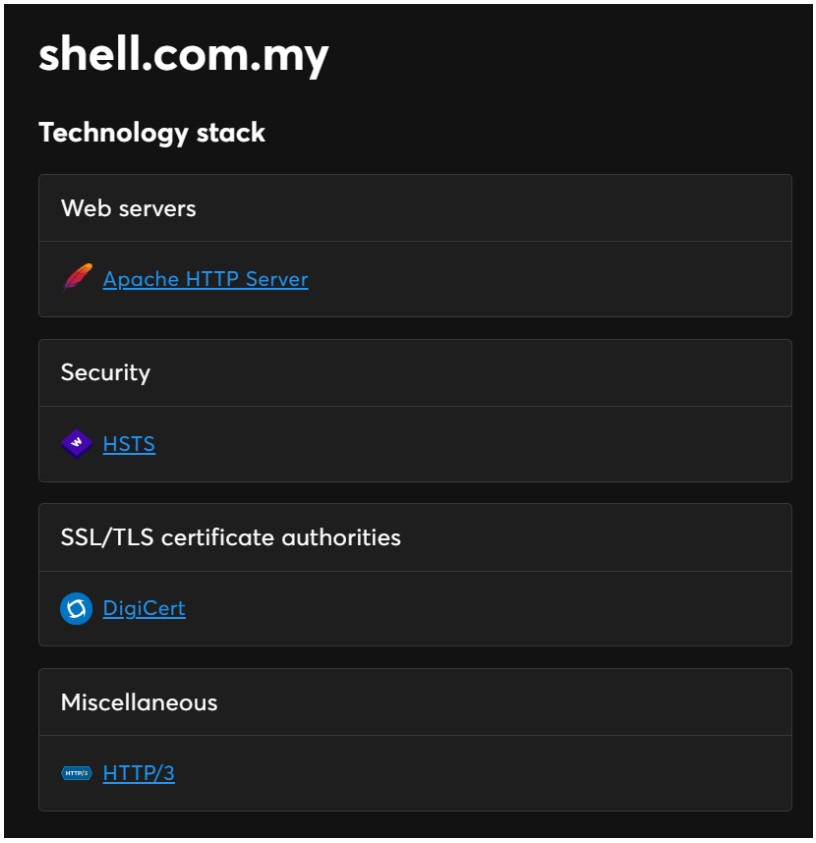
❖ Steps 2 : *Website Footprinting*

1) Whatweb

```
┌──(kali㉿kali)-[~]
└─$ whatweb https://www.shell.com.my
https://www.shell.com.my [403 Forbidden] Akamai-Global-Host, Country[UNITED STATES][US], HTTPServer[
ontent-type-options,server-timing]
```

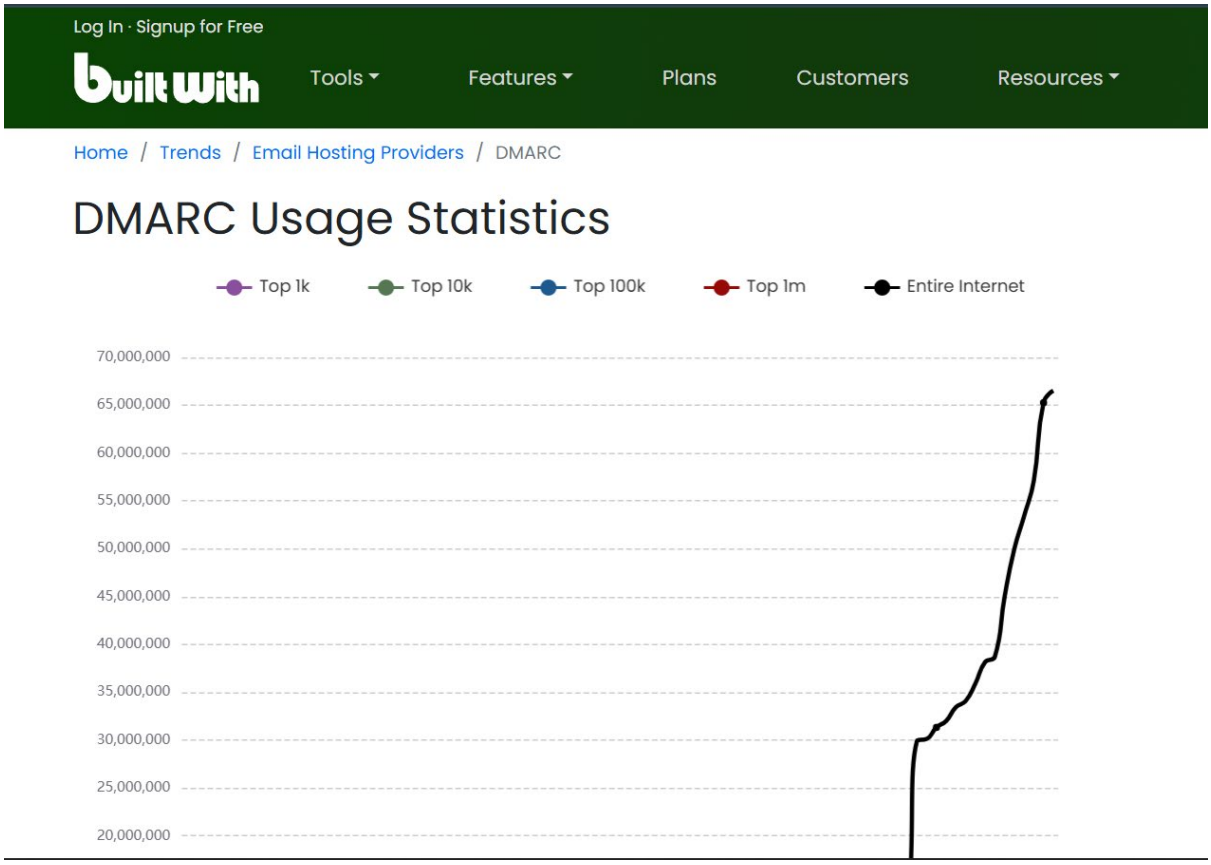
The whatweb tools identifies **shell.com.my** as **Akamai-Global-Host** indicate the use of Content Delivery Network (CDN) for security management. The request however shows a 403 Forbidden status, showing that the server is prohibited from automated access.

2) Wappalyzer



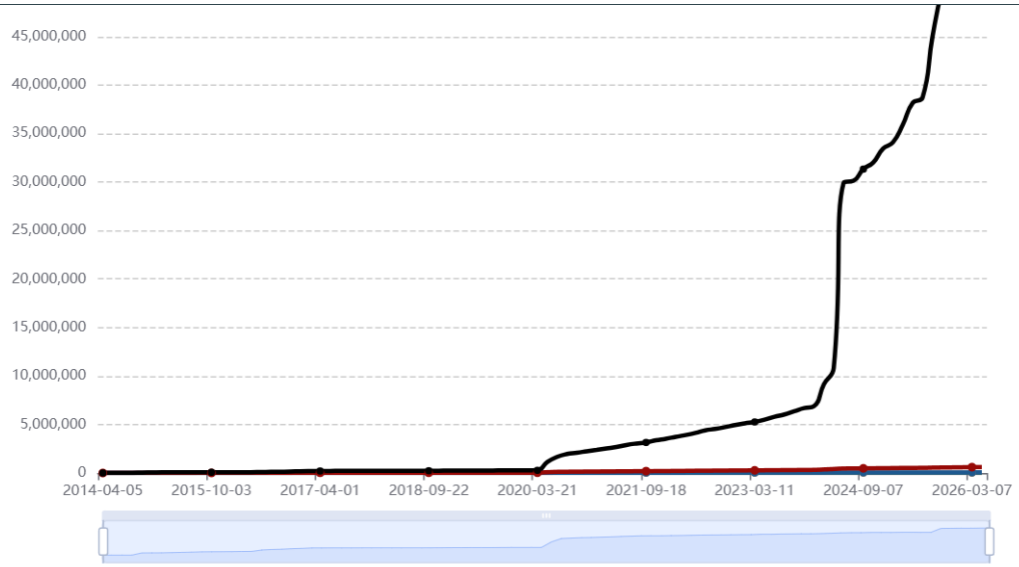
The web server uses **Apache HTTP** server, which consistent findings with the DNSDumpster. For security, it use **(HTTP Strict Transport Security)** to force secure HTTPS connections. Certificates are issued by high-assurance certificate authority.

3) Builtwith



This data is from BuiltWith which tracks the Shell's website technology. From the report, one of the found technology was DMARC (Domain-based Message Authentication, Reporting, and Conformance), which is a tools to reduce email abuse attacks.

The statistics shows a massive escalation trends in adopting it's into the website. This suggest that every large organizations should increasingly implement DMARC to protect their email from attacks, phishing, spoofing and ensure only authorized IP can send a message to their domain.



DMARC

A technical specification created by a group of organizations that want to help reduce the potential for email-based abuse

https://www.dmarc.org

Email Hosting Providers · DMARC

4) Netcraft

netcraft

LEARN MORE REPORT FRAUD

437 results (showing 1 to 20)

Rank	Site	First seen	Netblock	OS	Site Report
3748	www.rev**shell**s.com	May 2021	A100 ROW GmbH	Linux	
13057	www.**shell**y.com	May 1996	Shopify, Inc.	Linux	
18613	**shell**.cloud.google.com	October 2020	Google LLC	Linux	
20592	www.power**shell**gallery.com	July 2014	Microsoft Corporation	Linux	
21121	myaccount.**shell**pointmtg.com	December 2022	F5, Inc.	Linux	
21421	control.**shell**y.cloud	Febuary 2023	Google LLC	Linux	
26162	reopen**shell**y.cfd	January 2026	unknown	Linux	

The Netcraft keyword search for “shell” word to identify multiple website and target that host their domains, such as **Google LLC** and **Microsoft Corporation**. These shows most of the infrastructure actually runs on Linux os.

❖ Steps 3 : *Social Footprinting*

1) Social Media Search

The social searcher reveals Wael Sawan as the CEO of Shell, Investing RM 9 billion to Malaysia.

2) Sherlock

```
┌──(kali㉿kali)-[~]
└─$ sherlock Wael Sawan
Update available! 0.15.0 ──> 0.16.0
https://github.com/sherlock-project/sherlock/releases/tag/v0.16.0
[*] Checking username Wael on:

[+] 1337x: https://www.1337x.to/user/Wael/
[+] 7Cups: https://www.7cups.com/@Wael
[+] About.me: https://about.me/Wael
[+] Airliners: https://www.airliners.net/user/Wael/profile/photos
[+] Anilist: https://anilist.co/user/Wael/
[+] Apple Developer: https://developer.apple.com/forums/profile/Wael
[+] Apple Discussions: https://discussions.apple.com/profile/Wael
[+] Archive of Our Own: https://archiveofourown.org/users/Wael
[+] Archive.org: https://archive.org/details/@Wael
[+] Atcoder: https://atcoder.jp/users/Wael
[+] Vjudge: https://VJudge.net/user/Wael
[+] Audiojungle: https://audiojungle.net/user/Wael
[+] AWS Skills Profile: https://skillsprofile.skillbuilder.aws/user/Wael/
[+] Bandcamp: https://www.bandcamp.com/Wael
[+] Behance: https://www.behance.net/Wael
[+] Blipfoto: https://www.blipfoto.com/Wael
[+] Bluesky: https://bsky.app/profile/Wael.bsky.social
[+] BoardGameGeek: https://boardgamegeek.com/user/Wael
[+] BugCrowd: https://bugcrowd.com/Wael
[+] BuyMeACoffee: https://buymeacoff.ee/Wael
[+] BuzzFeed: https://buzzfeed.com/Wael
[+] CGTrader: https://www.cgtrader.com/Wael
[+] CashApp: https://cash.app/$Wael
[+] Chess: https://www.chess.com/member/Wael
[+] Clapper: https://clapperapp.com/Wael
[+] Clubhouse: https://www.clubhouse.com/@Wael
[+] Code Snippet Wiki: https://codesnippets.fandom.com/wiki/User:Wael
[+] Codeberg: https://codeberg.org/Wael
[+] Codecademy: https://www.codecademy.com/profiles/Wael
[+] Codechef: https://www.codechef.com/users/Wael
[+] Codeforces: https://codeforces.com/profile/Wael
[+] Coderwall: https://coderwall.com/Wael
[+] CodeSandbox: https://codesandbox.io/u/Wael
[+] Codolio: https://codolio.com/profile/Wael
[+] ColourLovers: https://www.colourlovers.com/lover/Wael
[+] Cplusplus: https://cplusplus.com/user/Wael
[+] Cracked Forum: https://cracked.sh/Wael
```

The Sherlock scan username “Wael Sawan” and find numerous matching profiles. The terminal output shows multiple potential account matches on **Apple Developer**, **Github**, **Codeacademy** and **Behance**. These results allow an analyst to gather the details in order to launch specific target by using spear-phishing attacks.

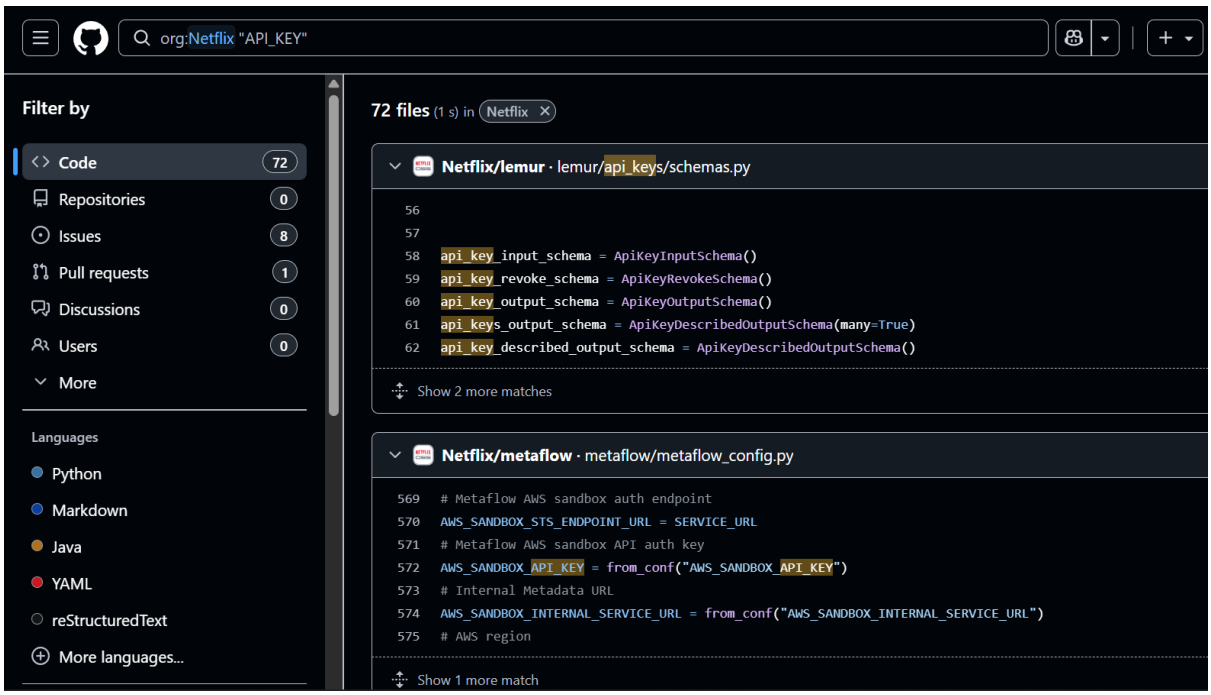
3) UserRecon Git

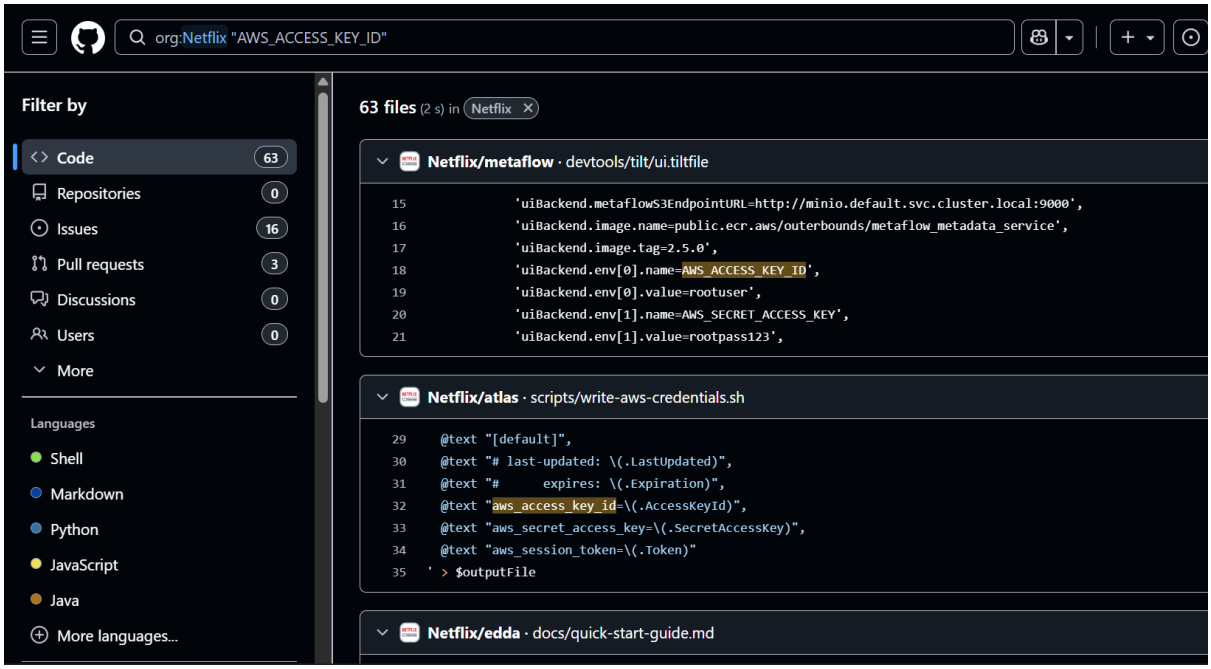
```
                                                    .-"""-.
                                                   /       \
                                                  /         \
                                                 //\       /\
                                                |\__\     /__/|
                                                 \    ||    /
                                                  \        /
                                                   \  __  /
                                                    '.__.'
 _   _                 ___
| | | |__   __ _ _ __ | _ \ __ __ __ _ _ __  //\
| | | / __|/ _ \ '_ \ |_) / _ v _/ _ \| '_ \ |\__\
| |_| \__ \  __/ | | |  _ <  __/ (_| (_) | | | \
 \___/|___/\___|_| |_| \_\___|\___\___/|_| |_|
                v1.0, Author: @thelinuxchoice

[?] Input Username: Wael Sawan

[*] Checking username Wael on:
[*] Checking username Sawan on:
[+] Instagram:  Found! https://www.instagram.com/Wael
 Found! https://www.instagram.com/Sawan
./userrecon.sh: line 53: $username.txt: ambiguous redirect
[+] Facebook:  Found! https://www.facebook.com/Wael
 Found! https://www.facebook.com/Sawan
./userrecon.sh: line 66: $username.txt: ambiguous redirect
[+] Twitter:  Found! https://www.twitter.com/Wael
 Found! https://www.twitter.com/Sawan
./userrecon.sh: line 79: $username.txt: ambiguous redirect
[+] YouTube:  Found! https://www.youtube.com/Wael
 Found! https://www.youtube.com/Sawan
./userrecon.sh: line 92: $username.txt: ambiguous redirect
[+] Blogger:  Found! https://Wael.blogspot.com
 Found! https://Sawan.blogspot.com
./userrecon.sh: line 105: $username.txt: ambiguous redirect
[+] GooglePlus:  Found! https://plus.google.com/+Wael/posts
 Found! https://plus.google.com/+Sawan/posts
./userrecon.sh: line 118: $username.txt: ambiguous redirect
[+] Reddit:  Found! https://www.reddit.com/user/Wael
 Found! https://www.reddit.com/user/Sawan
./userrecon.sh: line 133: $username.txt: ambiguous redirect
[+] Wordpress:  Found! https://Wael.wordpress.com
 Found! https://Sawan.wordpress.com
./userrecon.sh: line 146: $username.txt: ambiguous redirect
[+] Pinterest:  Found! https://www.pinterest.com/Wael
 Found! https://www.pinterest.com/Sawan
./userrecon.sh: line 159: $username.txt: ambiguous redirect
[+] GitHub:  Found! https://www.github.com/Wael
 Found! https://www.github.com/Sawan
./userrecon.sh: line 172: $username.txt: ambiguous redirect
[+] Tumblr:  Found! https://Wael.tumblr.com
 Found! https://Sawan.tumblr.com
./userrecon.sh: line 185: $username.txt: ambiguous redirect
[+] Flickr:  Found! https://www.flickr.com/photos/Wael
 Found! https://www.flickr.com/photos/Sawan
./userrecon.sh: line 198: $username.txt: ambiguous redirect
[+] Steam:  Found! https://steamcommunity.com/id/Wael
```

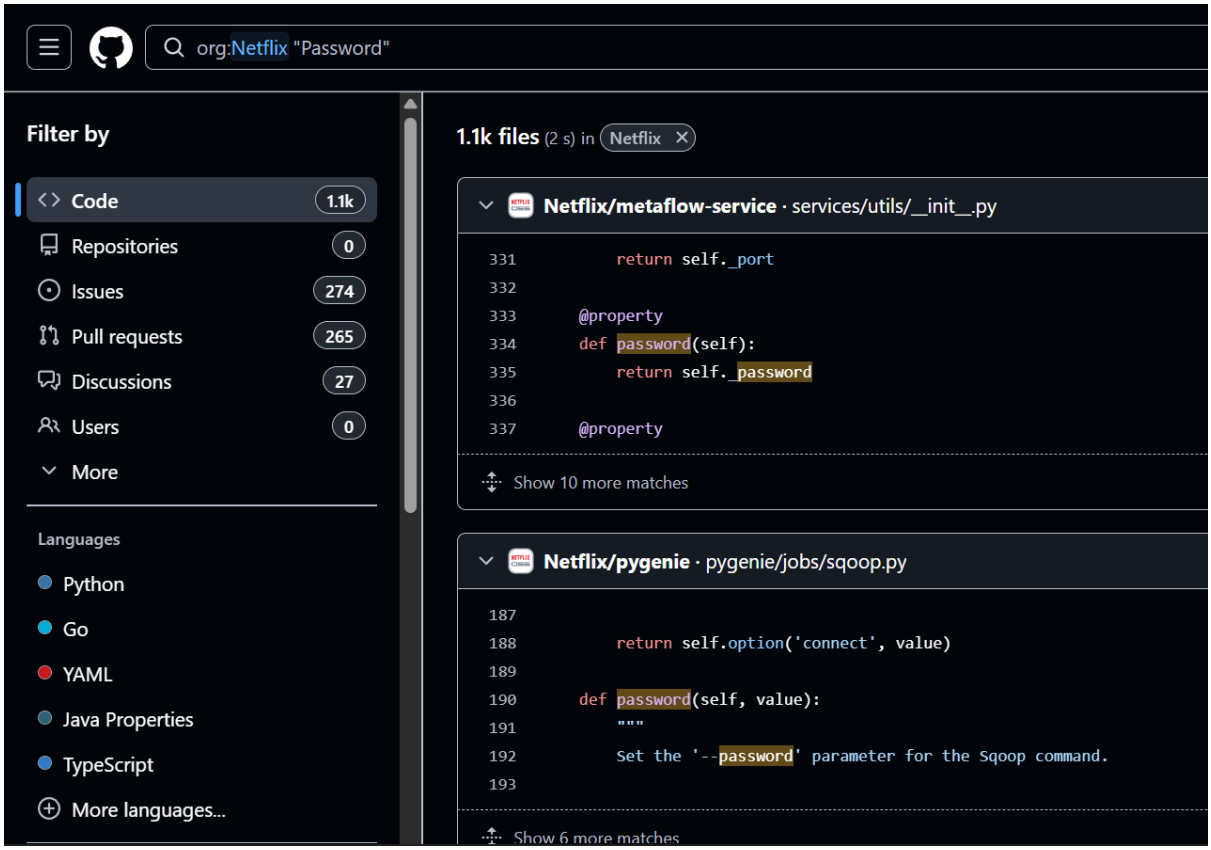
The UserRecon output displays the results of the scanning toward “Wael Sawan” which are the primary target. The script identified the potential matches on various sites, including **Instagram**, **Facebook**, **Twitter**, **Youtube** and **Github**.

❖ Steps 4 : *Source-Code Search*

1) GitDocker



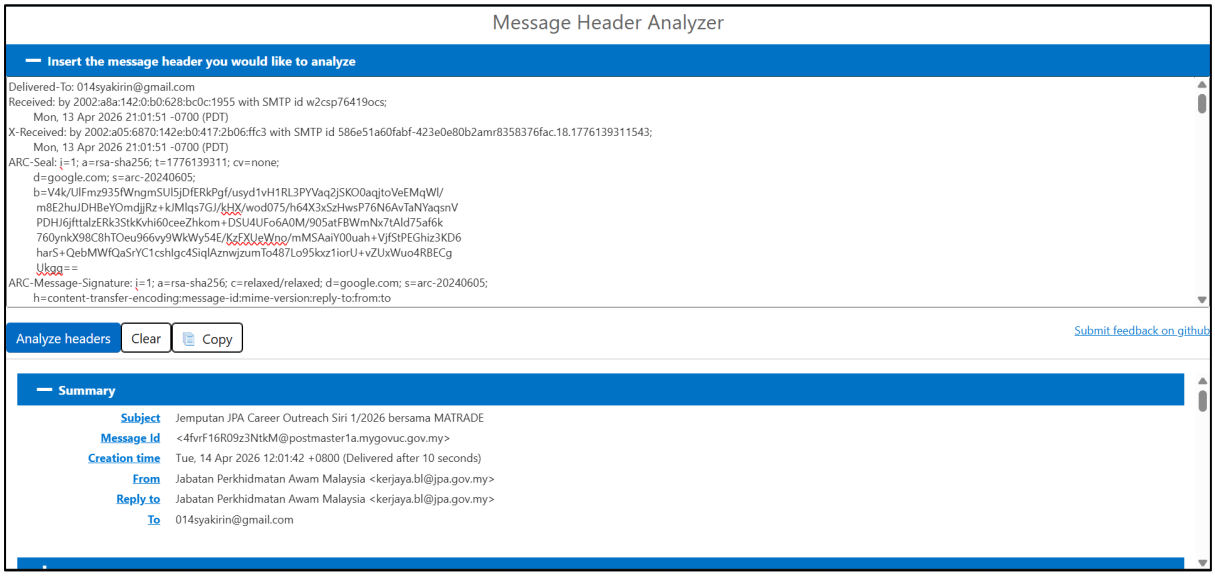




This method is called Github Dorking which are one of the way to search a specific source-code. It is a method that use a specific search operators and keyword list to locate specific information regarding sensitive credential which accidentally exposed in public repositories. By using a filtering for organization like Netflix and search terms for **“API_KEY”**, **“AWS_ACCESS_KEY_ID”**, or **“Password”**, we can identify a hardcoded credentials and configuration of the metadata.

❖ Steps 5 : *Email Footprinting*

1) Message Header Analyzer



Message Header Analyzer

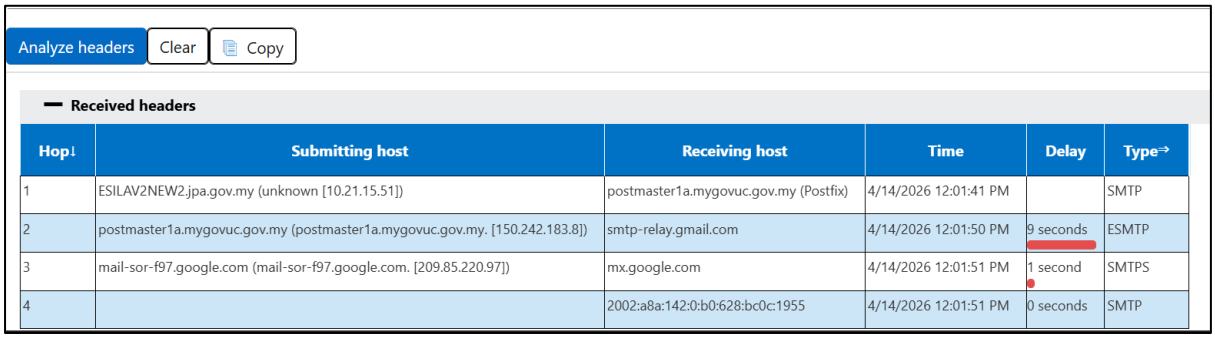
Insert the message header you would like to analyze

```
Delivered-To: 014syakirin@gmail.com
Received: by 2002:a8a:142:0:b0:628:bc0c:1955 with SMTP id w2csp76419ocs;
        Mon, 13 Apr 2026 21:01:51 -0700 (PDT)
X-Received: by 2002:a05:6870:142e:b0:417:2b06:ffc3 with SMTP id 586e51a60fabf-423e0e80b2amr8358376fac.18.1776139311543;
        Mon, 13 Apr 2026 21:01:51 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1776139311; cv=none;
        d=google.com; s=arc-20240605;
        b=V4k/UlFmz935fWngmSUI5jDfERkPgf/usyd1vH1RL3PYVaq2jSKO0aqjtoVeEMqWl/
         m8E2huJDHBeYOmdjjRz+kJMlqs7GJ/kHX/wod075/h64X3xSzHwsP76N6AvTaNYaqsnV
         PDHJ6jfttalzERk3StkKvhi60ceeZhkom+DSU4UFo6A0M/905atFBWmNx7tAld75af6k
         760ynkX98C8hTOeu966vy9WkWy54E/KzFXUeWnq/mMSAaiY00uah+VjfStPEGhiz3KD6
         harS+QebMWfQaSrYC1cshlgc4SiqlAznwjzumTo487Lo95kxz1iorU+vZUxWuo4RBECg
         Ukqq==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20240605;
        h=content-transfer-encoding:message-id:mime-version:reply-to:from:to
```

Analyze headers | Clear | Copy | Submit feedback on github

Summary

Subject	Jemputan JPA Career Outreach Siri 1/2026 bersama MATRADE
Message Id	<4fvrF16R09z3NtkM@postmaster1a.mygovuc.gov.my>
Creation time	Tue, 14 Apr 2026 12:01:42 +0800 (Delivered after 10 seconds)
From	Jabatan Perkhidmatan Awam Malaysia <kerjaya.bl@jpa.gov.my>
Reply to	Jabatan Perkhidmatan Awam Malaysia <kerjaya.bl@jpa.gov.my>
To	014syakirin@gmail.com



Analyze headers | Clear | Copy

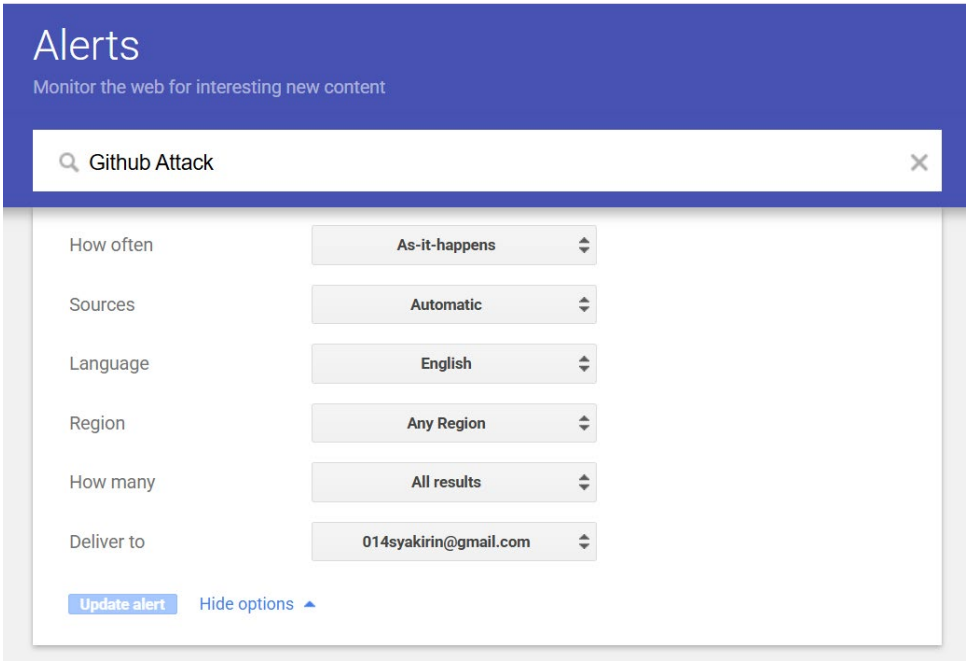
Received headers

Hop	Submitting host	Receiving host	Time	Delay	Type
1	ESILAV2NEW2.jpa.gov.my (unknown [10.21.15.51])	postmaster1a.mygovuc.gov.my (Postfix)	4/14/2026 12:01:41 PM		SMTP
2	postmaster1a.mygovuc.gov.my (postmaster1a.mygovuc.gov.my. [150.242.183.8])	smtp-relay.gmail.com	4/14/2026 12:01:50 PM	9 seconds	ESMTP
3	mail-sor-f97.google.com (mail-sor-f97.google.com. [209.85.220.97])	mx.google.com	4/14/2026 12:01:51 PM	1 second	SMTPS
4		2002:a8a:142:0:b0:628:bc0c:1955	4/14/2026 12:01:51 PM	0 seconds	SMTP

The Message Header Analyzer provides a detailed of the email sender, or the path it travelled in the internet. By analyzing these received headers, we successfully obtain the originated IP address **10.21.15.51** in first hop and next servers IP which is **150.242.183.8**. The delay for the email to reach each server is between 9 second occurred from **mygovuc.gow.my** to Google's relay. The analyzer identifies the used protocols which are **ESMTP** and **SMTPS**, directly proof the email was transmitted via a secure channels.

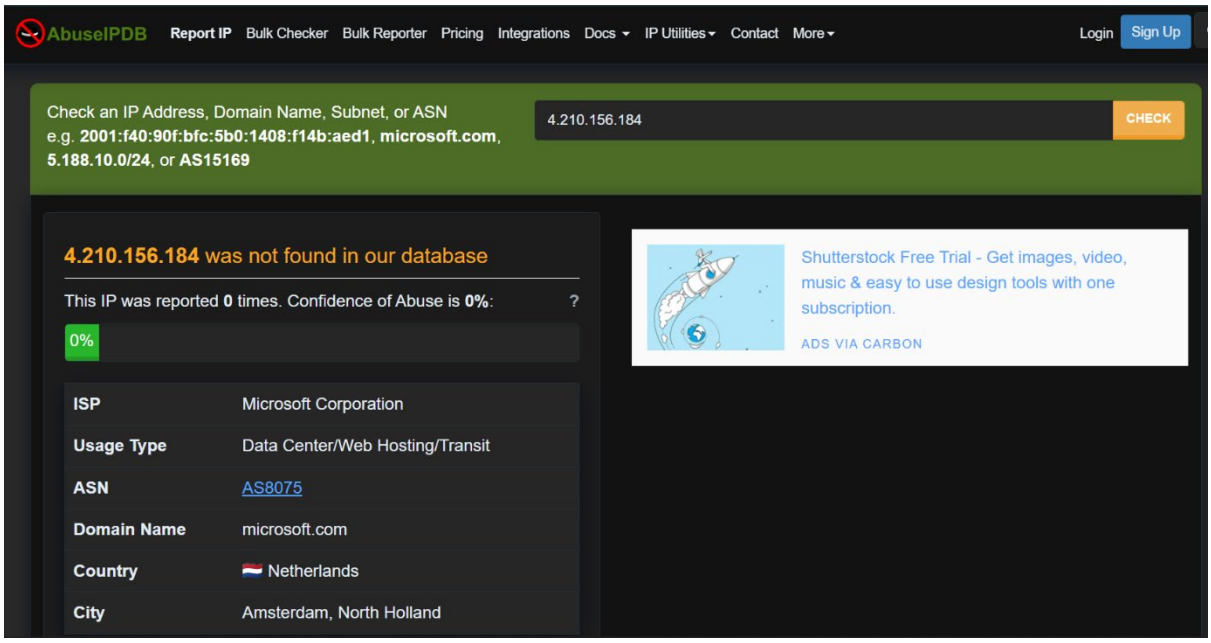
❖ Steps 6 : *Reputation and Alerts*

1) Google Alert



Setting up **Google Alert** for issues like **"Github Attack"** provides a regular monitoring to track online mentions of any threat related. By configuring the alert for "As-it-happens" from automatic sources, we can ensure a real-time global information gathering.

2) AbuseIPDB



The **AbuseIPDB** results for the resolved IP **4.210.156.184** provide a certified verification of the target reputation. The scan shows 0 reports and a 0% Confidence of Abuse, which directly saying this IP is not associated with malicious activity.

Summary

Step	Tool	Finding	Interpretation	Mitigation
Domain Footprinting	MYNIC WHOIS	Registered 1997, DNSSEC signed	A high-reputation domain with default protection against DNS hijacking	Use domain lock and monitor expiry dates
	nslookup	IP : 4.210.156.184	Target using Microsoft Azure infrastructure	Use WAF/CDN to mask true original IP
	DNSDumpster	Have multiple digital infrastructure in US, Netherlands, Ireland and Malaysia	Using Akamai CDNs for global distribution	Segment the internal vs public assets
Website Footprinting	whatweb	403 Forbidden, Akamai Global Host	Automated scanning is prohibited, traffic managed by Akamai group	Remove version banners and patch regularly
	Wappalyzer	Use Apache server: HSTS and HTTP/3	A modern web standards usage, forcing a secure connections	Keep CMS and plugins updated
Social Footprinting	Sherlock	Have potential matches on Instagram, Apple Discussion and About me	Potential for spear-phishing based on CEO social media	Employee awareness training
	Github Dorking	Hardcoded keys and metadata founded	Found a leakage of API keys and internal configurations	Scan for secret and use git hooks
Email Footprinting	Header Analyzer	Hop delays and identification of IP address	Detailed mail route maps from JPA server	Enforce MFA and DMARC policies
Reputation and Monitoring	Google Alerts	Monitor for “Github Attack”	Real-time tracking of evolving threat	Reset password after the breach alerts
	AbuseIPDB	0 reports, 0% of abuse	Verify this domains a clean Azure server IP	Monitoring this IP for future blacklisting